
EECS 16B Designing Information Devices and Systems II

Discussion Worksheet

Discussion 13B

1. Quadratic Approximation and Vector Differentiation

As shown in the previous discussion, a common way to approximate a non-linear high-dimensional functions is to perform linearization near a point. In the case of a two-dimensional function $f(x, y)$ with scalar output, the linear approximation of $f(x, y)$ at a point (x_*, y_*) is given by

$$f(x, y) \approx f(x_*, y_*) + f_x(x_*, y_*)(x - x_*) + f_y(x_*, y_*)(y - y_*) \quad (1)$$

where as in the previous section,

$$f_x(x_*, y_*) = \left. \frac{\partial f(x, y)}{\partial x} \right|_{(x_*, y_*)} \quad \text{and} \quad f_y(x_*, y_*) = \left. \frac{\partial f(x, y)}{\partial y} \right|_{(x_*, y_*)}. \quad (2)$$

In vector form, this can be written as:

$$f(\vec{x}) \approx f(\vec{x}_*) + \left[D_{\vec{x}} f|_{\vec{x}_*} \right] (\vec{x} - \vec{x}_*). \quad (3)$$

Recall from the previous discussion that $D_{\vec{x}} f$ is a row-vector filled with the partial derivatives $\frac{\partial f(\vec{x})}{\partial x_i}$:

$$D_{\vec{x}} f = \left[\frac{\partial f(\vec{x})}{\partial x_1} \quad \dots \quad \frac{\partial f(\vec{x})}{\partial x_n} \right] = \left[f_{x_1}(\vec{x}) \quad \dots \quad f_{x_n}(\vec{x}) \right]. \quad (4)$$

Our goal is to extend this idea to a quadratic approximation. To do this, we need some notion of a second derivative.

For this discussion, we will only be considering these types of functions from $\mathbb{R}^n \rightarrow \mathbb{R}$, since that is the typical form for a cost function used during optimization.

- (a) Given the function $f(x) = e^{-2x}$, find the first and second derivatives, and write out its quadratic approximation at $x = x_*$.

Hint: Use Taylor's theorem.

- (b) To write second partial derivatives compactly, we will introduce a new notation that builds off the notation f_x and f_y introduced previously. To compute f_{xy} , we first take the derivative in x , then in y :

$$f_{xy}(x_*, y_*) = \left. \frac{\partial f_x(x, y)}{\partial y} \right|_{(x_*, y_*)} = \left. \frac{\partial^2 f(x, y)}{\partial y \partial x} \right|_{(x_*, y_*)}. \quad (5)$$

Given the function $f(x, y) = x^2 y^2$, find all of the first and second partial derivatives.

- (c) To find the quadratic approximation of $f(x, y)$ near (x_*, y_*) , we plug in $f(x_* + \Delta x, y_* + \Delta y)$ and drop the terms that are higher order than quadratic:

$$f(x_* + \Delta x, y_* + \Delta y) = (x_* + \Delta x)^2 (y_* + \Delta y)^2 \quad (6)$$

$$= (x_*^2 + 2x_*\Delta x + (\Delta x)^2)(y_*^2 + 2y_*\Delta y + (\Delta y)^2) \quad (7)$$

$$\approx x_*^2 y_*^2 + 2x_* y_*^2 \Delta x + 2x_*^2 y_* \Delta y \quad (8)$$

$$+ y_*^2 (\Delta x)^2 + 4x_* y_* (\Delta x)(\Delta y) + x_*^2 (\Delta y)^2 \quad (9)$$

$$= f(x_*, y_*) + f_x(x_*, y_*)\Delta x + f_y(x_*, y_*)\Delta y \quad (10)$$

$$+ \frac{1}{2} f_{xx}(x_*, y_*) (\Delta x)^2 + \frac{1}{2} f_{yy}(x_*, y_*) (\Delta y)^2 \quad (11)$$

$$+ f_{xy}(x_*, y_*) (\Delta x)(\Delta y). \quad (12)$$

This is slightly different from the expression we get via the Taylor series expansion. How would we rewrite this expression, so that *all* second derivatives are involved, each with a coefficient of $\frac{1}{2}$?

- (d) Just as we created the derivative row vector to hold all the first partial derivatives to help in writing linearization in matrix/vector form:

$$D_{\vec{x}}f = \left[\frac{\partial f(\vec{x})}{\partial x_1} \quad \cdots \quad \frac{\partial f(\vec{x})}{\partial x_n} \right] = \left[f_{x_1}(\vec{x}) \quad \cdots \quad f_{x_n}(\vec{x}) \right] \quad (13)$$

we would like to create a matrix to hold all the second partial derivatives to help in writing quadratic approximation in matrix/vector form:

$$H_{\vec{x}}f = \begin{bmatrix} \frac{\partial^2 f(\vec{x})}{\partial x_1^2} & \cdots & \frac{\partial^2 f(\vec{x})}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\vec{x})}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(\vec{x})}{\partial x_n^2} \end{bmatrix} = \begin{bmatrix} f_{x_1 x_1}(\vec{x}) & \cdots & f_{x_n x_1}(\vec{x}) \\ \vdots & \ddots & \vdots \\ f_{x_1 x_n}(\vec{x}) & \cdots & f_{x_n x_n}(\vec{x}) \end{bmatrix} \quad (14)$$

This matrix is the *Hessian* of f . Note that this quantity is different from the *Jacobian* matrix that was covered in the previous discussion. In contrast to the Hessian, which is the matrix of second partial derivatives of a *scalar-valued vector-input* function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, the Jacobian is the matrix of first partial derivatives of a *vector-valued vector-input* function $\vec{f}: \mathbb{R}^n \rightarrow \mathbb{R}^k$.

In fact, the Hessian is the (Jacobian) derivative of the derivative; if we let $\vec{g}(\vec{x}) = (D_{\vec{x}}f)^\top$ (so that it's a column vector and the dimensions work out), then $H_{\vec{x}}f = D_{\vec{x}}\vec{g}$.

To get a feel for the Hessian of f , find $H_{(x,y)}f$ for the f above, that is, $f(x, y) = x^2 y^2$.

- (e) Using the Hessian, write out the general formula for the quadratic approximation of a scalar-valued function f of a vector $\vec{x}$ in vector/matrix form.

- (f) Show that the quadratic approximation for the scalar-valued function $f(\vec{w}) = e^{\vec{x}^\top \vec{w}}$ around $\vec{w} = \vec{w}_\star$ is

$$f(\vec{w}_\star + \Delta \vec{w}) \approx e^{\vec{x}^\top \vec{w}_\star} \left(1 + \vec{x}^\top (\Delta \vec{w}) + \frac{1}{2} \left(\vec{x}^\top (\Delta \vec{w}) \right)^2 \right). \quad (15)$$

Here, assume that $\vec{x}$ is just some given vector — a constant vector.

Hint: You can compute the following partial derivatives:

$$f_{w_i}(\vec{w}) = x_i f(\vec{w}) \quad (16)$$

$$f_{w_i w_j}(\vec{w}) = x_i x_j f(\vec{w}). \quad (17)$$

Now compute $D_{\vec{w}} f$ and $H_{\vec{w}} f$, and plug it into the quadratic approximation formula.

- (g) Use linearity to give the quadratic approximation for the function $\sum_{i=1}^m e^{\vec{x}_i^\top \vec{w}}$ around $\vec{w} = \vec{w}_\star$. Here, assume that the $\vec{x}_i$ are just some given vectors.

- (h) **Practice.** The second derivative also has an interpretation as the derivative of the derivative. However, we saw that the derivative of a scalar-valued function with respect to a vector is naturally a row. If you

wanted to approximate how much the derivative changed by moving a small amount $\Delta \vec{w}$, how would you get such an estimate using your expression for the second derivative?

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